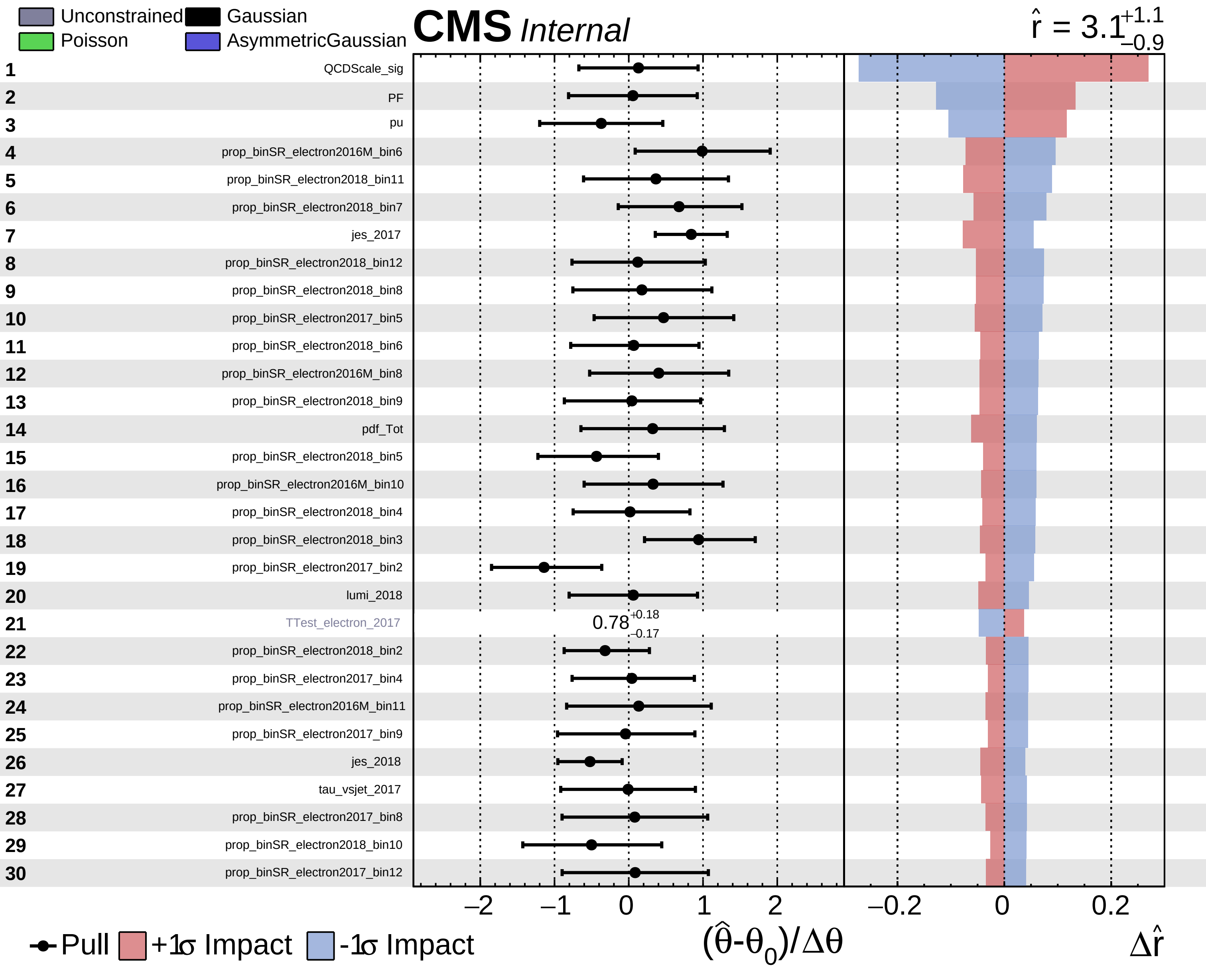
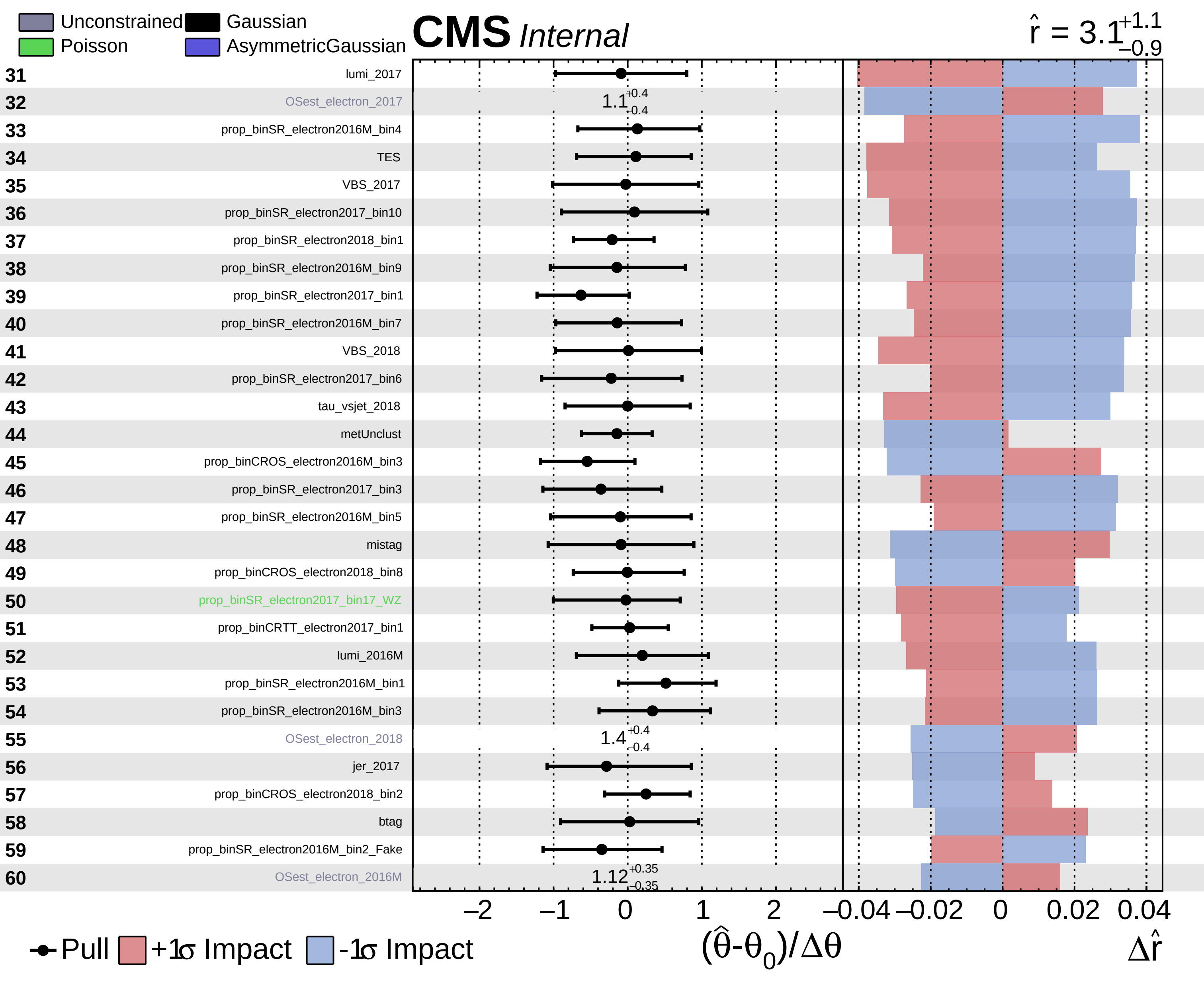
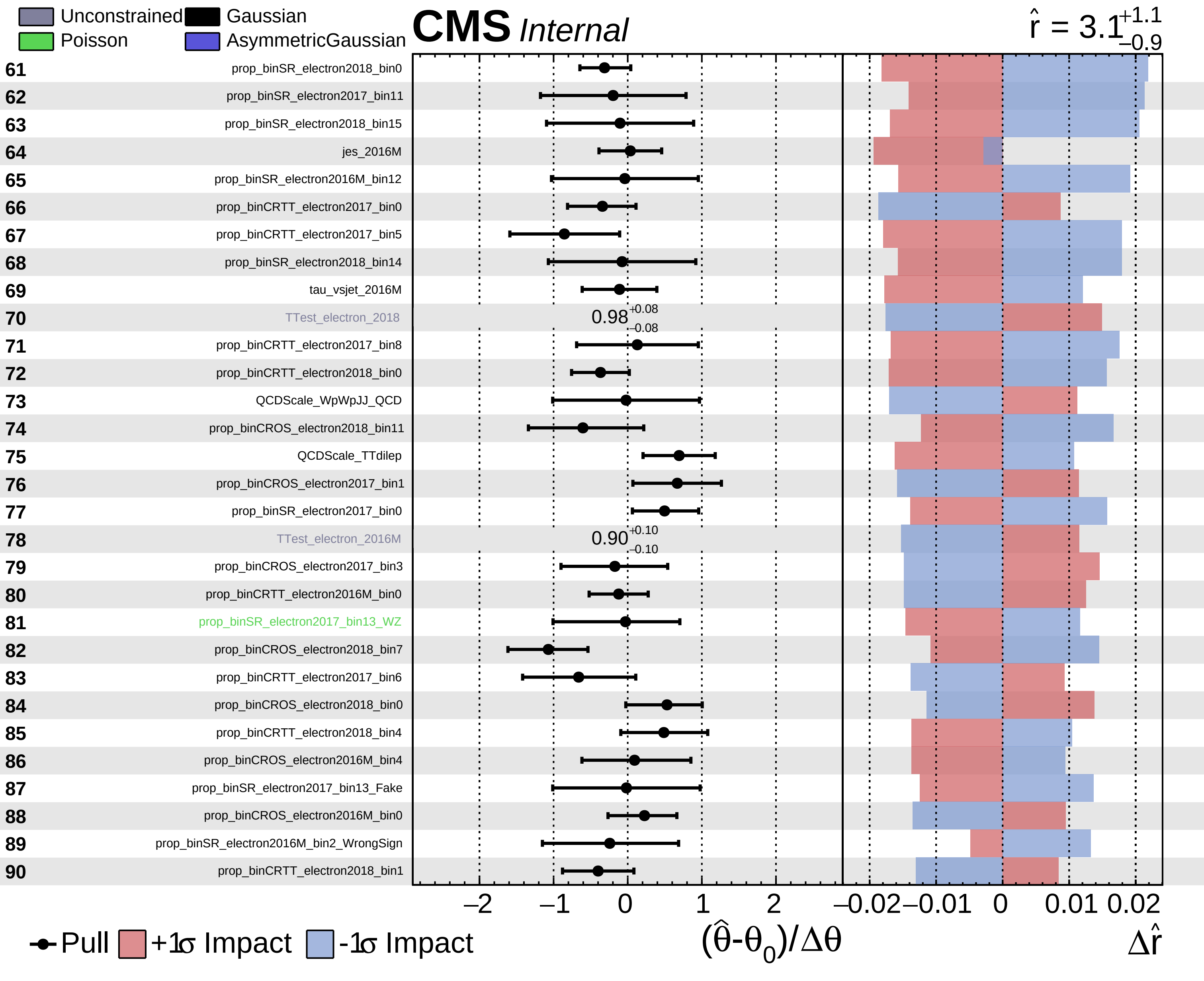
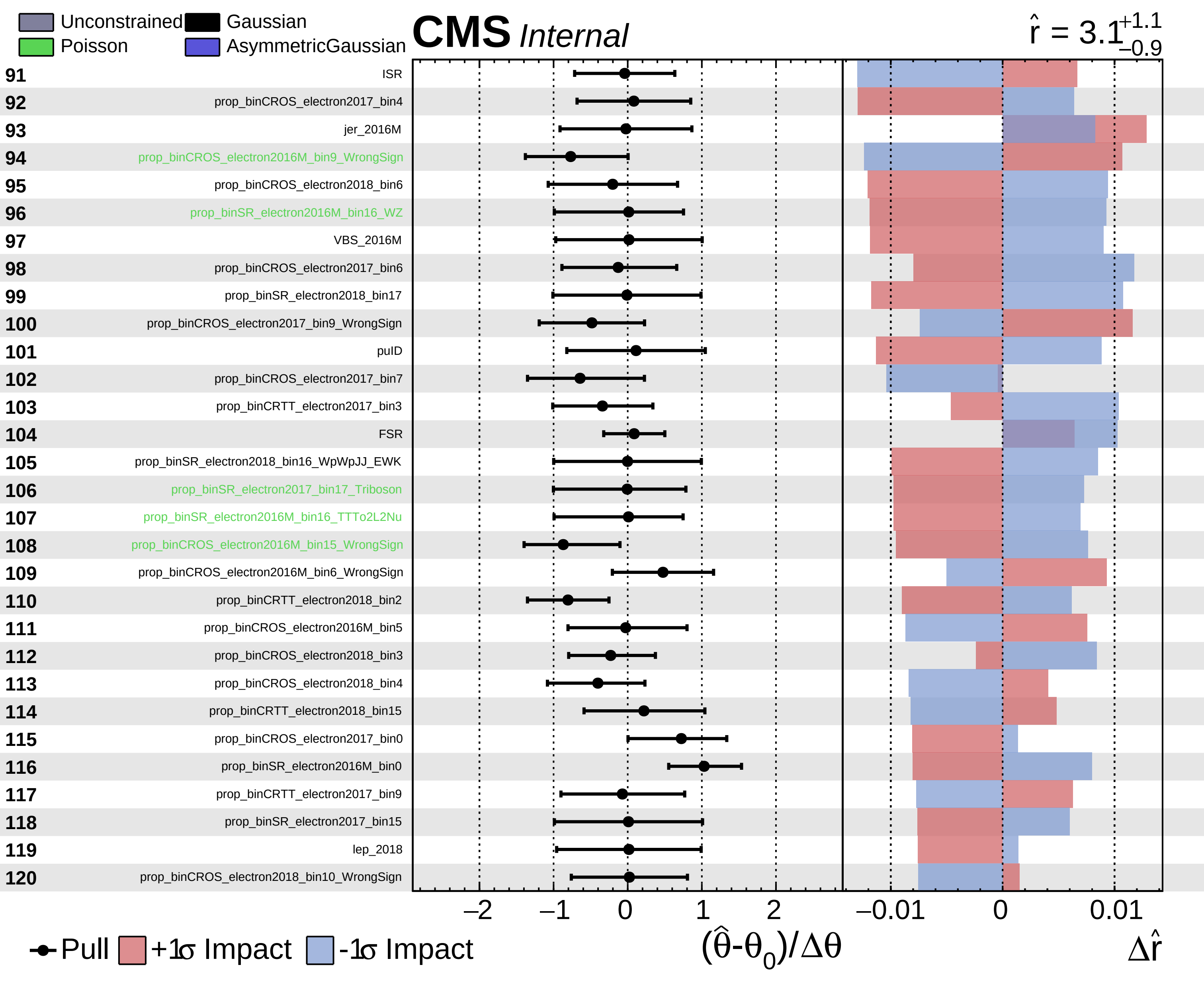

Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
r̂ = 3.1+1.1 −0.9
1 QCDScale_sig
2 PF
3 pu
4 prop_binSR_electron2016M_bin6
5 prop_binSR_electron2018_bin11
6 prop_binSR_electron2018_bin7
7 jes_2017
8 prop_binSR_electron2018_bin12
9 prop_binSR_electron2018_bin8
10 prop_binSR_electron2017_bin5
11 prop_binSR_electron2018_bin6
12 prop_binSR_electron2016M_bin8
13 prop_binSR_electron2018_bin9
14 pdf_Tot
15 prop_binSR_electron2018_bin5
16 prop_binSR_electron2016M_bin10
17 prop_binSR_electron2018_bin4
18 prop_binSR_electron2018_bin3
19 prop_binSR_electron2017_bin2
20 lumi_2018
21 TTest_electron_2017
0.78+0.18 −0.17
22 prop_binSR_electron2018_bin2
23 prop_binSR_electron2017_bin4
24 prop_binSR_electron2016M_bin11
25 prop_binSR_electron2017_bin9
26 jes_2018
27 tau_vsjet_2017
28 prop_binSR_electron2017_bin8
29 prop_binSR_electron2018_bin10
30 prop_binSR_electron2017_bin12
−2 −1 0 1 2
−0.2 0 0.2
Pull
+1σ Impact
-1σ Impact
(θ̂-θ0)/Δθ
Δr̂

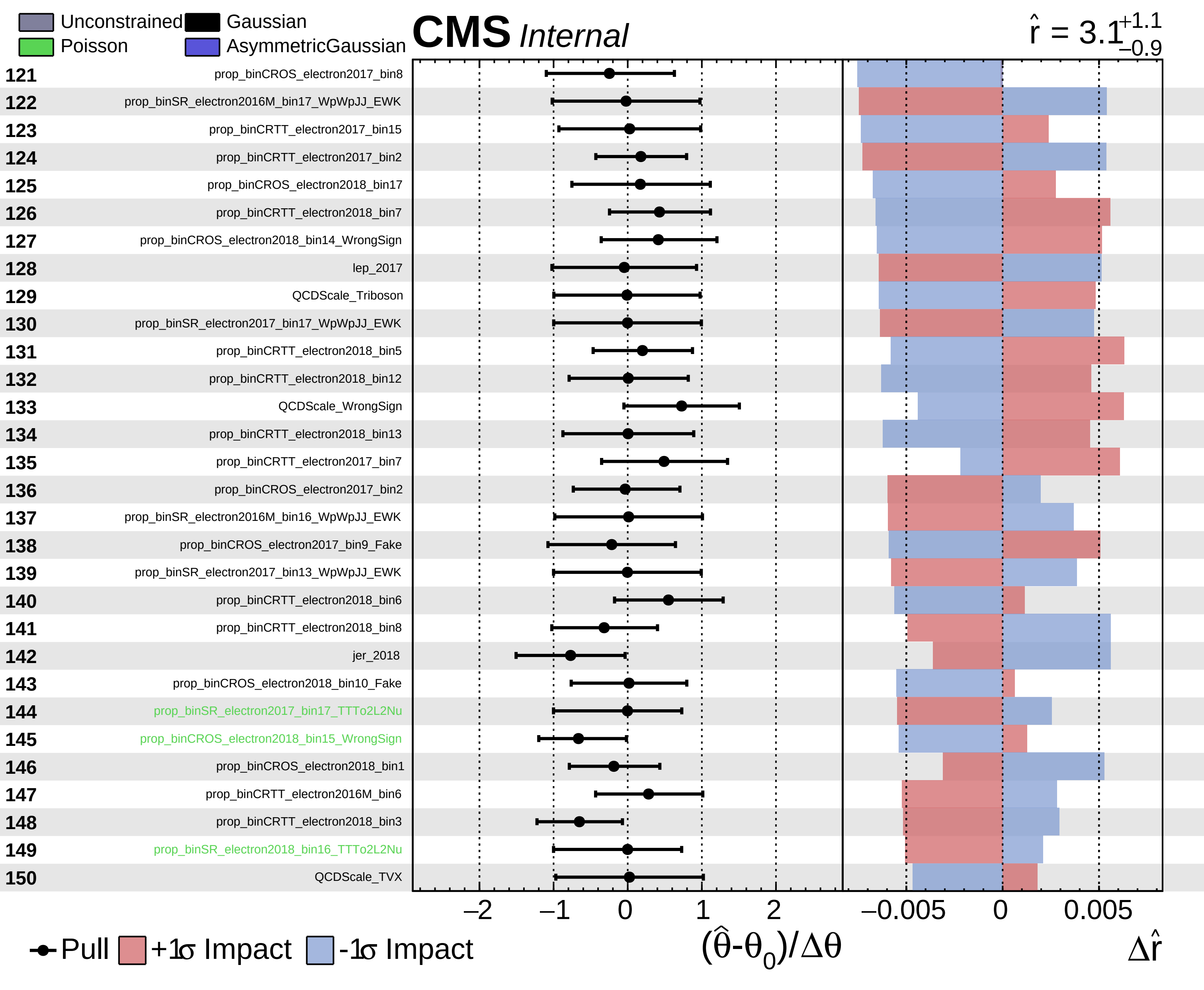


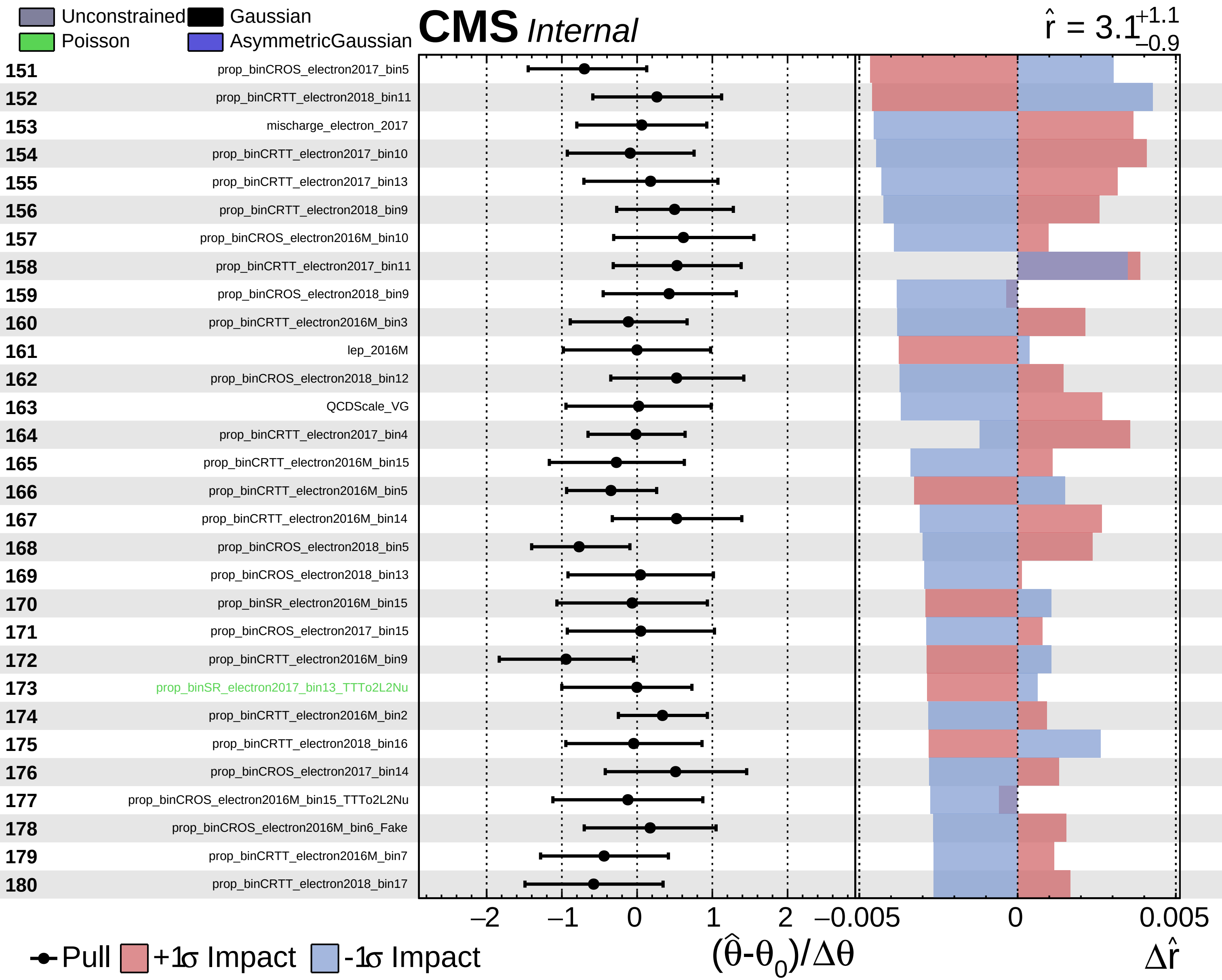
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
r̂ = 3.1 +1.1 −0.9
31 lumi_2017
32 OSest_electron_2017 1.1 +0.4 −0.4
33 prop_binSR_electron2016M_bin4
34 TES
35 VBS_2017
36 prop_binSR_electron2017_bin10
37 prop_binSR_electron2018_bin1
38 prop_binSR_electron2016M_bin9
39 prop_binSR_electron2017_bin1
40 prop_binSR_electron2016M_bin7
41 VBS_2018
42 prop_binSR_electron2017_bin6
43 tau_vsjet_2018
44 metUnclust
45 prop_binCROS_electron2016M_bin3
46 prop_binSR_electron2017_bin3
47 prop_binSR_electron2016M_bin5
48 mistag
49 prop_binCROS_electron2018_bin8
50 prop_binSR_electron2017_bin17_WZ
51 prop_binCRTT_electron2017_bin1
52 lumi_2016M
53 prop_binSR_electron2016M_bin1
54 prop_binSR_electron2016M_bin3
55 OSest_electron_2018 1.4 +0.4 −0.4
56 jer_2017
57 prop_binCROS_electron2018_bin2
58 btag
59 prop_binSR_electron2016M_bin2_Fake
60 OSest_electron_2016M 1.12 +0.35 −0.35
−2 −1 0 1 2
−0.04 −0.02 0 0.02 0.04
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$



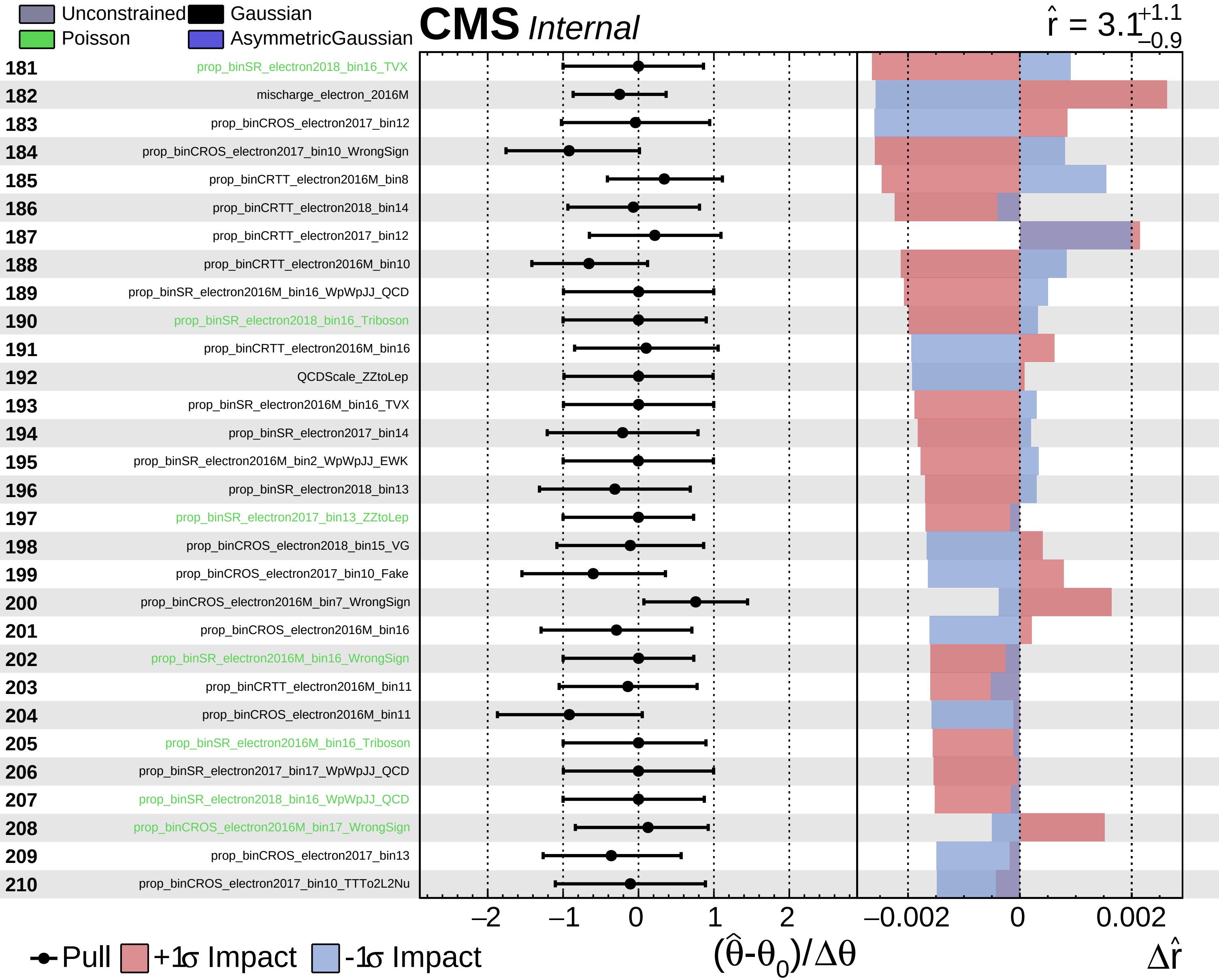
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 3.1^{+1.1}_{-0.9}$
61 prop_binSR_electron2018_bin0
62 prop_binSR_electron2017_bin11
63 prop_binSR_electron2018_bin15
64 jes_2016M
65 prop_binSR_electron2016M_bin12
66 prop_binCRTT_electron2017_bin0
67 prop_binCRTT_electron2017_bin5
68 prop_binSR_electron2018_bin14
69 tau_vsjet_2016M
70 TTest_electron_2018
$0.98^{+0.08}_{-0.08}$
71 prop_binCRTT_electron2017_bin8
72 prop_binCRTT_electron2018_bin0
73 QCDScale_WpWpJJ_QCD
74 prop_binCROS_electron2018_bin11
75 QCDScale_TTdilep
76 prop_binCROS_electron2017_bin1
77 prop_binSR_electron2017_bin0
78 TTest_electron_2016M
$0.90^{+0.10}_{-0.10}$
79 prop_binCROS_electron2017_bin3
80 prop_binCRTT_electron2016M_bin0
81 prop_binSR_electron2017_bin13_WZ
82 prop_binCROS_electron2018_bin7
83 prop_binCRTT_electron2017_bin6
84 prop_binCROS_electron2018_bin0
85 prop_binCRTT_electron2018_bin4
86 prop_binCROS_electron2016M_bin4
87 prop_binSR_electron2017_bin13_Fake
88 prop_binCROS_electron2016M_bin0
89 prop_binSR_electron2016M_bin2_WrongSign
90 prop_binCRTT_electron2018_bin1
−2 −1 0 1 2
−0.02 −0.01 0 0.01 0.02
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 3.1^{+1.1}_{-0.9}$
91 ISR
92 prop_binCROS_electron2017_bin4
93 jer_2016M
94 prop_binCROS_electron2016M_bin9_WrongSign
95 prop_binCROS_electron2018_bin6
96 prop_binSR_electron2016M_bin16_WZ
97 VBS_2016M
98 prop_binCROS_electron2017_bin6
99 prop_binSR_electron2018_bin17
100 prop_binCROS_electron2017_bin9_WrongSign
101 puID
102 prop_binCROS_electron2017_bin7
103 prop_binCRTT_electron2017_bin3
104 FSR
105 prop_binSR_electron2018_bin16_WpWpJJ_EWK
106 prop_binSR_electron2017_bin17_Triboson
107 prop_binSR_electron2016M_bin16_TTTo2L2Nu
108 prop_binCROS_electron2016M_bin15_WrongSign
109 prop_binCROS_electron2016M_bin6_WrongSign
110 prop_binCRTT_electron2018_bin2
111 prop_binCROS_electron2016M_bin5
112 prop_binCROS_electron2018_bin3
113 prop_binCROS_electron2018_bin4
114 prop_binCRTT_electron2018_bin15
115 prop_binCROS_electron2017_bin0
116 prop_binSR_electron2016M_bin0
117 prop_binCRTT_electron2017_bin9
118 prop_binSR_electron2017_bin15
119 lep_2018
120 prop_binCROS_electron2018_bin10_WrongSign
−2 −1 0 1 2
−0.01 0 0.01
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$

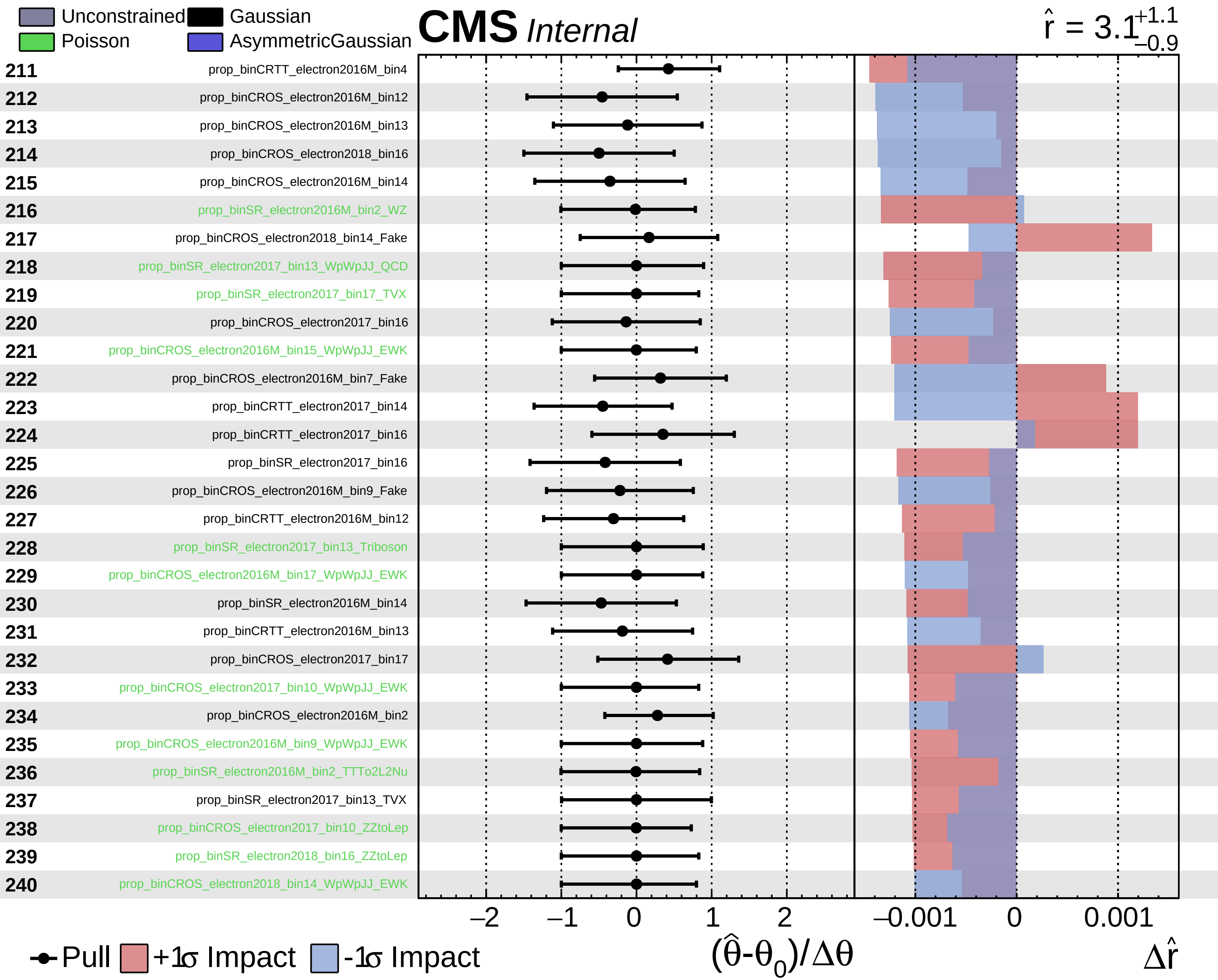

Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 3.1^{+1.1}_{-0.9}$
121 prop_binCROS_electron2017_bin8
122 prop_binSR_electron2016M_bin17_WpWpJJ_EWK
123 prop_binCRTT_electron2017_bin15
124 prop_binCRTT_electron2017_bin2
125 prop_binCROS_electron2018_bin17
126 prop_binCRTT_electron2018_bin7
127 prop_binCROS_electron2018_bin14_WrongSign
128 lep_2017
129 QCDScale_Triboson
130 prop_binSR_electron2017_bin17_WpWpJJ_EWK
131 prop_binCRTT_electron2018_bin5
132 prop_binCRTT_electron2018_bin12
133 QCDScale_WrongSign
134 prop_binCRTT_electron2018_bin13
135 prop_binCRTT_electron2017_bin7
136 prop_binCROS_electron2017_bin2
137 prop_binSR_electron2016M_bin16_WpWpJJ_EWK
138 prop_binCROS_electron2017_bin9_Fake
139 prop_binSR_electron2017_bin13_WpWpJJ_EWK
140 prop_binCRTT_electron2018_bin6
141 prop_binCRTT_electron2018_bin8
142 jer_2018
143 prop_binCROS_electron2018_bin10_Fake
144 prop_binSR_electron2017_bin17_TTTo2L2Nu
145 prop_binCROS_electron2018_bin15_WrongSign
146 prop_binCROS_electron2018_bin1
147 prop_binCRTT_electron2016M_bin6
148 prop_binCRTT_electron2018_bin3
149 prop_binSR_electron2018_bin16_TTTo2L2Nu
150 QCDScale_TVX
−2
−1
0
1
2
−0.005
0
0.005
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$



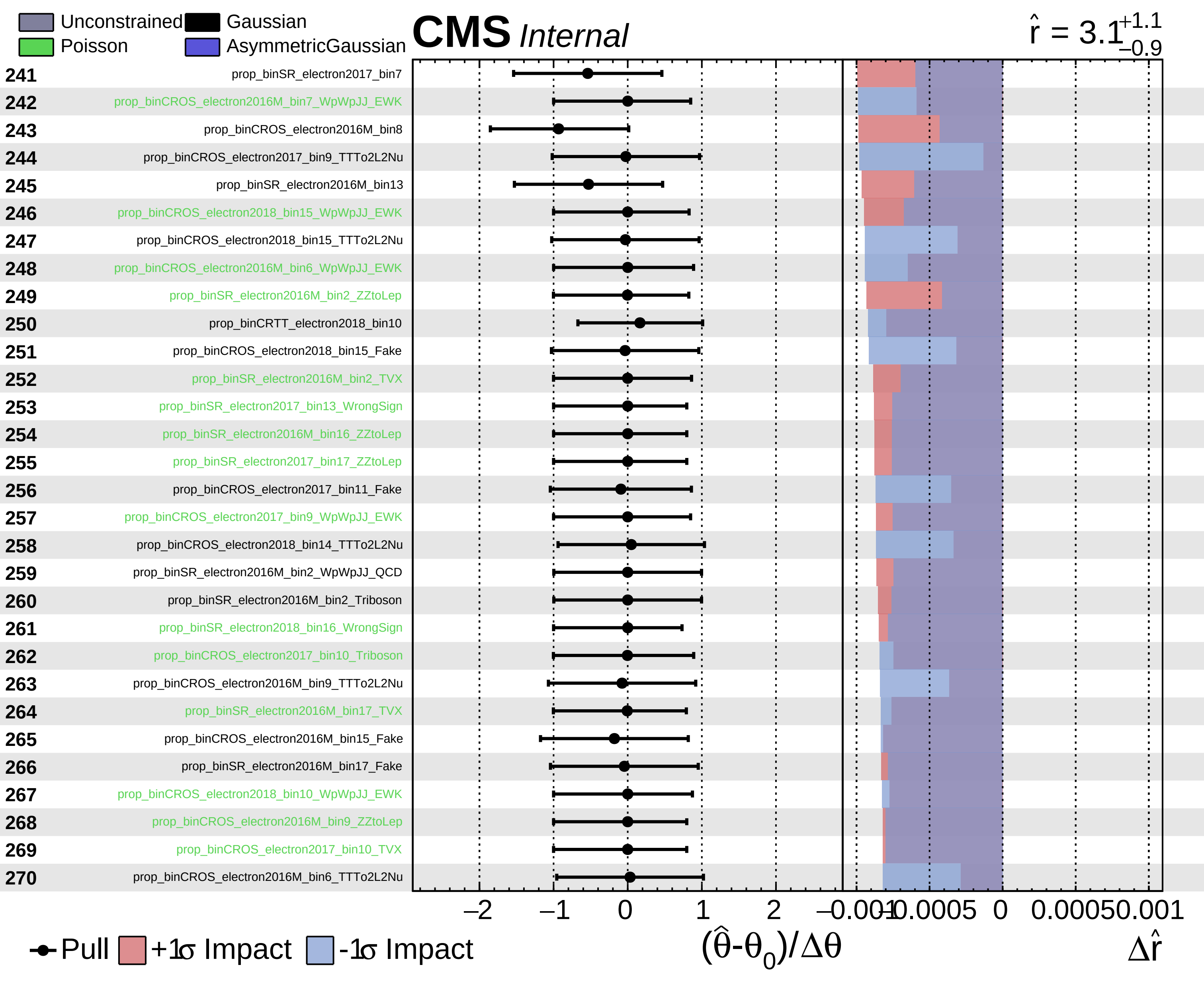
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 3.1^{+1.1}_{-0.9}$
151 prop_binCROS_electron2017_bin5
152 prop_binCRTT_electron2018_bin11
153 mischarge_electron_2017
154 prop_binCRTT_electron2017_bin10
155 prop_binCRTT_electron2017_bin13
156 prop_binCRTT_electron2018_bin9
157 prop_binCROS_electron2016M_bin10
158 prop_binCRTT_electron2017_bin11
159 prop_binCROS_electron2018_bin9
160 prop_binCRTT_electron2016M_bin3
161 lep_2016M
162 prop_binCROS_electron2018_bin12
163 QCDScale_VG
164 prop_binCRTT_electron2017_bin4
165 prop_binCRTT_electron2016M_bin15
166 prop_binCRTT_electron2016M_bin5
167 prop_binCRTT_electron2016M_bin14
168 prop_binCROS_electron2018_bin5
169 prop_binCROS_electron2018_bin13
170 prop_binSR_electron2016M_bin15
171 prop_binCROS_electron2017_bin15
172 prop_binCRTT_electron2016M_bin9
173 prop_binSR_electron2017_bin13_TTTo2L2Nu
174 prop_binCRTT_electron2016M_bin2
175 prop_binCRTT_electron2018_bin16
176 prop_binCROS_electron2017_bin14
177 prop_binCROS_electron2016M_bin15_TTTo2L2Nu
178 prop_binCROS_electron2016M_bin6_Fake
179 prop_binCRTT_electron2016M_bin7
180 prop_binCRTT_electron2018_bin17
−2
−1
0
1
2
−0.005
0
0.005
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$



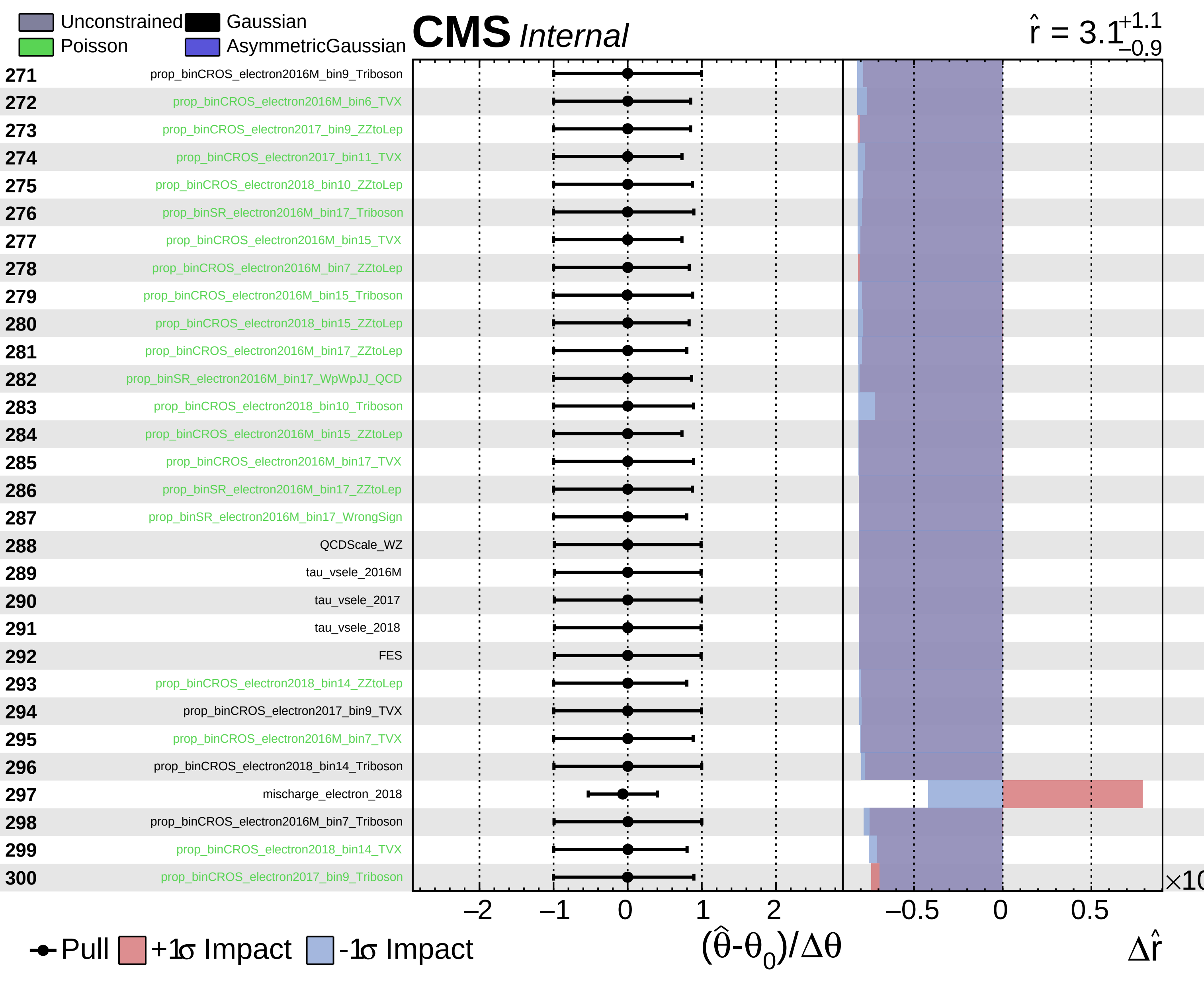
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
r̂ = 3.1 +1.1 −0.9
181 prop_binSR_electron2018_bin16_TVX
182 mischarge_electron_2016M
183 prop_binCROS_electron2017_bin12
184 prop_binCROS_electron2017_bin10_WrongSign
185 prop_binCRTT_electron2016M_bin8
186 prop_binCRTT_electron2018_bin14
187 prop_binCRTT_electron2017_bin12
188 prop_binCRTT_electron2016M_bin10
189 prop_binSR_electron2016M_bin16_WpWpJJ_QCD
190 prop_binSR_electron2018_bin16_Triboson
191 prop_binCRTT_electron2016M_bin16
192 QCDScale_ZZtoLep
193 prop_binSR_electron2016M_bin16_TVX
194 prop_binSR_electron2017_bin14
195 prop_binSR_electron2016M_bin2_WpWpJJ_EWK
196 prop_binSR_electron2018_bin13
197 prop_binSR_electron2017_bin13_ZZtoLep
198 prop_binCROS_electron2018_bin15_VG
199 prop_binCROS_electron2017_bin10_Fake
200 prop_binCROS_electron2016M_bin7_WrongSign
201 prop_binCROS_electron2016M_bin16
202 prop_binSR_electron2016M_bin16_WrongSign
203 prop_binCRTT_electron2016M_bin11
204 prop_binCROS_electron2016M_bin11
205 prop_binSR_electron2016M_bin16_Triboson
206 prop_binSR_electron2017_bin17_WpWpJJ_QCD
207 prop_binSR_electron2018_bin16_WpWpJJ_QCD
208 prop_binCROS_electron2016M_bin17_WrongSign
209 prop_binCROS_electron2017_bin13
210 prop_binCROS_electron2017_bin10_TTTo2L2Nu
−2 −1 0 1 2
−0.002 0 0.002
Pull
+1σ Impact
-1σ Impact
(θ̂-θ0)/Δθ
Δr̂



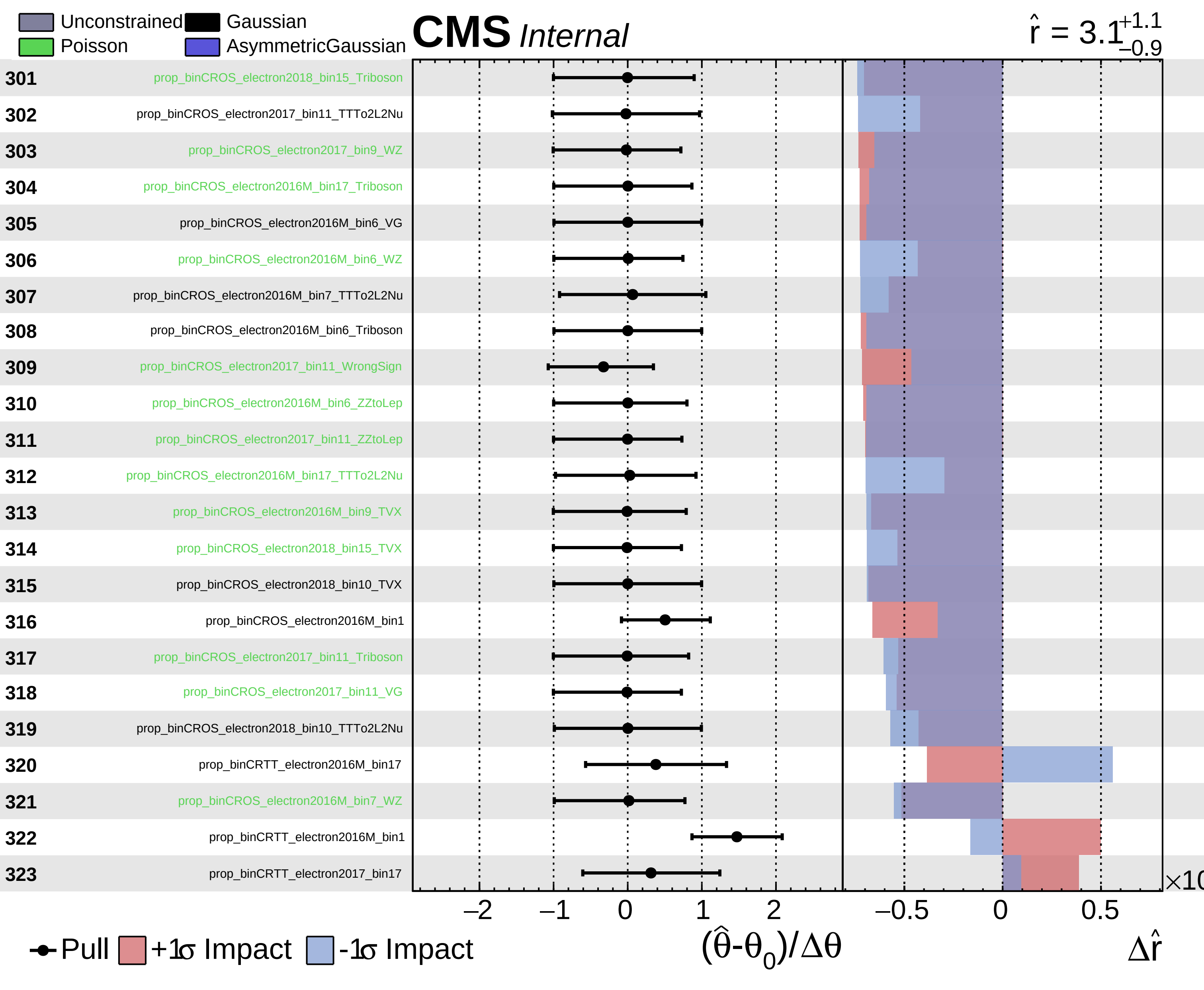
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 3.1^{+1.1}_{-0.9}$
211 prop_binCRTT_electron2016M_bin4
212 prop_binCROS_electron2016M_bin12
213 prop_binCROS_electron2016M_bin13
214 prop_binCROS_electron2018_bin16
215 prop_binCROS_electron2016M_bin14
216 prop_binSR_electron2016M_bin2_WZ
217 prop_binCROS_electron2018_bin14_Fake
218 prop_binSR_electron2017_bin13_WpWpJJ_QCD
219 prop_binSR_electron2017_bin17_TVX
220 prop_binCROS_electron2017_bin16
221 prop_binCROS_electron2016M_bin15_WpWpJJ_EWK
222 prop_binCROS_electron2016M_bin7_Fake
223 prop_binCRTT_electron2017_bin14
224 prop_binCRTT_electron2017_bin16
225 prop_binSR_electron2017_bin16
226 prop_binCROS_electron2016M_bin9_Fake
227 prop_binCRTT_electron2016M_bin12
228 prop_binSR_electron2017_bin13_Triboson
229 prop_binCROS_electron2016M_bin17_WpWpJJ_EWK
230 prop_binSR_electron2016M_bin14
231 prop_binCRTT_electron2016M_bin13
232 prop_binCROS_electron2017_bin17
233 prop_binCROS_electron2017_bin10_WpWpJJ_EWK
234 prop_binCROS_electron2016M_bin2
235 prop_binCROS_electron2016M_bin9_WpWpJJ_EWK
236 prop_binSR_electron2016M_bin2_TTTo2L2Nu
237 prop_binSR_electron2017_bin13_TVX
238 prop_binCROS_electron2017_bin10_ZZtoLep
239 prop_binSR_electron2018_bin16_ZZtoLep
240 prop_binCROS_electron2018_bin14_WpWpJJ_EWK
−2
−1
0
1
2
−0.001
0
0.001
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$



Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 3.1^{+1.1}_{-0.9}$
241 prop_binSR_electron2017_bin7
242 prop_binCROS_electron2016M_bin7_WpWpJJ_EWK
243 prop_binCROS_electron2016M_bin8
244 prop_binCROS_electron2017_bin9_TTTo2L2Nu
245 prop_binSR_electron2016M_bin13
246 prop_binCROS_electron2018_bin15_WpWpJJ_EWK
247 prop_binCROS_electron2018_bin15_TTTo2L2Nu
248 prop_binCROS_electron2016M_bin6_WpWpJJ_EWK
249 prop_binSR_electron2016M_bin2_ZZtoLep
250 prop_binCRTT_electron2018_bin10
251 prop_binCROS_electron2018_bin15_Fake
252 prop_binSR_electron2016M_bin2_TVX
253 prop_binSR_electron2017_bin13_WrongSign
254 prop_binSR_electron2016M_bin16_ZZtoLep
255 prop_binSR_electron2017_bin17_ZZtoLep
256 prop_binCROS_electron2017_bin11_Fake
257 prop_binCROS_electron2017_bin9_WpWpJJ_EWK
258 prop_binCROS_electron2018_bin14_TTTo2L2Nu
259 prop_binSR_electron2016M_bin2_WpWpJJ_QCD
260 prop_binSR_electron2016M_bin2_Triboson
261 prop_binSR_electron2018_bin16_WrongSign
262 prop_binCROS_electron2017_bin10_Triboson
263 prop_binCROS_electron2016M_bin9_TTTo2L2Nu
264 prop_binSR_electron2016M_bin17_TVX
265 prop_binCROS_electron2016M_bin15_Fake
266 prop_binSR_electron2016M_bin17_Fake
267 prop_binCROS_electron2018_bin10_WpWpJJ_EWK
268 prop_binCROS_electron2016M_bin9_ZZtoLep
269 prop_binCROS_electron2017_bin10_TVX
270 prop_binCROS_electron2016M_bin6_TTTo2L2Nu
−2
−1
0
1
2
−0.001
−0.0005
0
0.0005
0.001
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$



Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 3.1^{+1.1}_{-0.9}$
271 prop_binCROS_electron2016M_bin9_Triboson
272 prop_binCROS_electron2016M_bin6_TVX
273 prop_binCROS_electron2017_bin9_ZZtoLep
274 prop_binCROS_electron2017_bin11_TVX
275 prop_binCROS_electron2018_bin10_ZZtoLep
276 prop_binSR_electron2016M_bin17_Triboson
277 prop_binCROS_electron2016M_bin15_TVX
278 prop_binCROS_electron2016M_bin7_ZZtoLep
279 prop_binCROS_electron2016M_bin15_Triboson
280 prop_binCROS_electron2018_bin15_ZZtoLep
281 prop_binCROS_electron2016M_bin17_ZZtoLep
282 prop_binSR_electron2016M_bin17_WpWpJJ_QCD
283 prop_binCROS_electron2018_bin10_Triboson
284 prop_binCROS_electron2016M_bin15_ZZtoLep
285 prop_binCROS_electron2016M_bin17_TVX
286 prop_binSR_electron2016M_bin17_ZZtoLep
287 prop_binSR_electron2016M_bin17_WrongSign
288 QCDScale_WZ
289 tau_vsele_2016M
290 tau_vsele_2017
291 tau_vsele_2018
292 FES
293 prop_binCROS_electron2018_bin14_ZZtoLep
294 prop_binCROS_electron2017_bin9_TVX
295 prop_binCROS_electron2016M_bin7_TVX
296 prop_binCROS_electron2018_bin14_Triboson
297 mischarge_electron_2018
298 prop_binCROS_electron2016M_bin7_Triboson
299 prop_binCROS_electron2018_bin14_TVX
300 prop_binCROS_electron2017_bin9_Triboson
−2
−1
0
1
2
−0.5
0
0.5
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$



Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 3.1^{+1.1}_{-0.9}$
301 prop_binCROS_electron2018_bin15_Triboson
302 prop_binCROS_electron2017_bin11_TTTo2L2Nu
303 prop_binCROS_electron2017_bin9_WZ
304 prop_binCROS_electron2016M_bin17_Triboson
305 prop_binCROS_electron2016M_bin6_VG
306 prop_binCROS_electron2016M_bin6_WZ
307 prop_binCROS_electron2016M_bin7_TTTo2L2Nu
308 prop_binCROS_electron2016M_bin6_Triboson
309 prop_binCROS_electron2017_bin11_WrongSign
310 prop_binCROS_electron2016M_bin6_ZZtoLep
311 prop_binCROS_electron2017_bin11_ZZtoLep
312 prop_binCROS_electron2016M_bin17_TTTo2L2Nu
313 prop_binCROS_electron2016M_bin9_TVX
314 prop_binCROS_electron2018_bin15_TVX
315 prop_binCROS_electron2018_bin10_TVX
316 prop_binCROS_electron2016M_bin1
317 prop_binCROS_electron2017_bin11_Triboson
318 prop_binCROS_electron2017_bin11_VG
319 prop_binCROS_electron2018_bin10_TTTo2L2Nu
320 prop_binCRTT_electron2016M_bin17
321 prop_binCROS_electron2016M_bin7_WZ
322 prop_binCRTT_electron2016M_bin1
323 prop_binCRTT_electron2017_bin17
−2 −1 0 1 2
−0.5 0 0.5
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$